

Review Questions - Week 2

Q1 State the relationship between the inner product of two vectors u, v , the angle θ between them, and their norms $\|u\|, \|v\|$.

Q2 State the Cauchy-Schwarz inequality (the proof has not yet been covered in the course).

Q3 State and prove the Triangle Inequality.

Q4 State and prove the Pythagorean theorem.

Q5 Define the matrix-vector product. Provide three different ways to express or interpret it.

Q6 Determine the coefficient matrix and the augmented matrix for the following system:

$$\begin{cases} 3x_1 + x_2 + 2x_3 = 0 \\ x_1 + 3x_3 = 2 \\ x_1 + 4x_2 = -1 \end{cases}$$

Q7 Given two vectors $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^n$, list three different notations used to represent their inner product.

Q8 For a matrix A , describe the notation used for its individual entries. Express A in terms of its row vectors and in terms of its column vectors.

Q9 What is an overdetermined system of linear equations? What is an underdetermined system? What is a square system?

Q10 Define triangular (upper and lower) and diagonal matrices using mathematical notation regarding their entries.

Q11 Solve the linear system of equations represented by the following augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 6 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 3 & 10 \end{array} \right]$$

Q12 List the three elementary row operations allowed in Gauss-Jordan elimination.